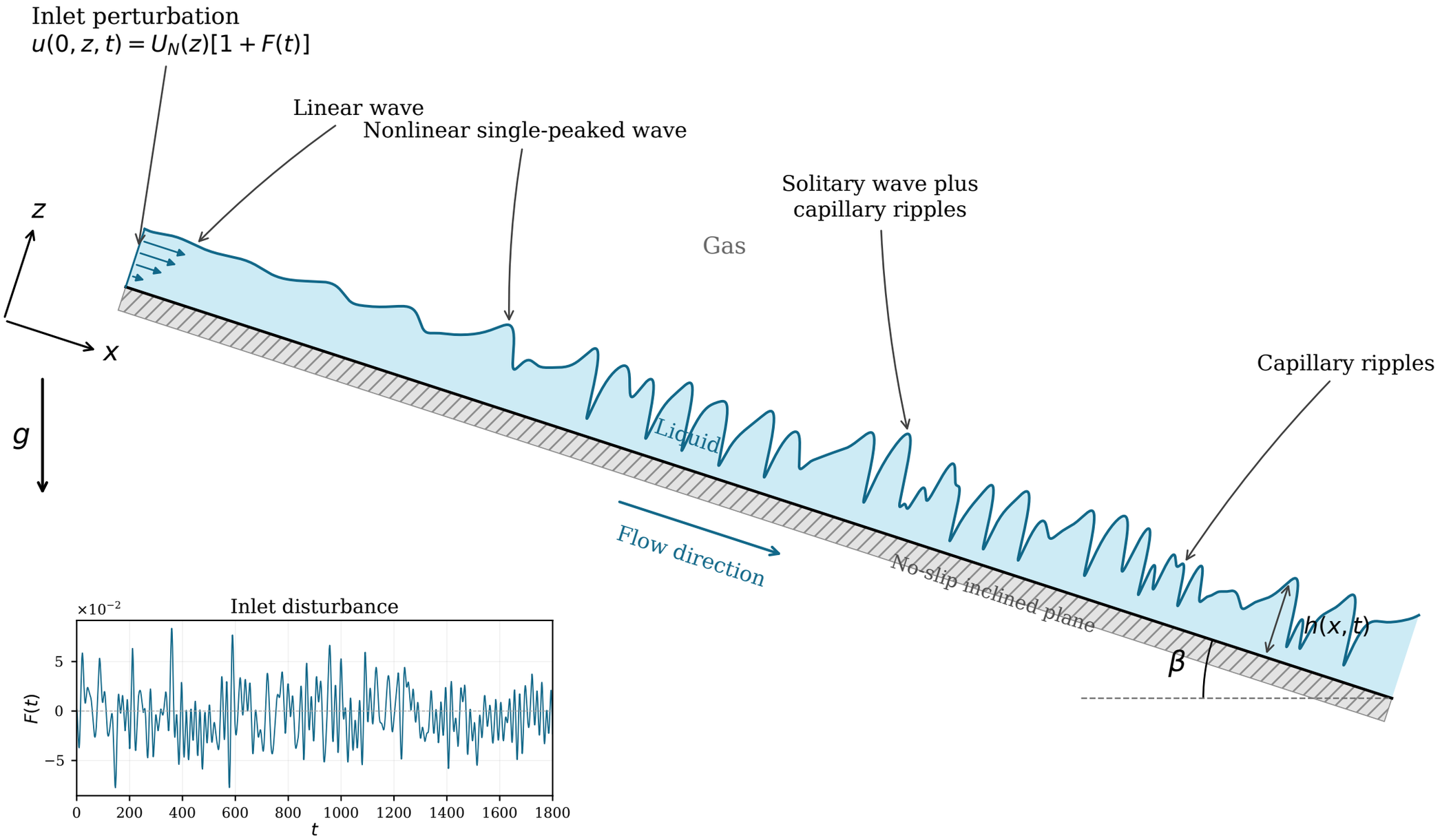

Inlet perturbation
$u(0,z,t) = U_N(z)[1+F(t)]$
Linear wave
Nonlinear single-peaked wave
Solitary wave plus
capillary ripples
Capillary ripples
Gas
Liquid
Flow direction
No-slip inclined plane
$h(x,t)$
β
z
x
g
Inlet disturbance
$\times 10^{-2}$
$F(t)$
t
5
0
−5
0
200
400
600
800
1000
1200
1400
1600
1800